

A Replication of Poppels & Kehler Experiment 3 (2019)

1 Introduction

Verb phrase ellipsis is a sentence construction in which the verb is omitted from one clause of the sentence, but the meaning is still recoverable through context. The reduction in sentence acceptability when this ellipsis clause and its antecedent differ in voice is referred to as Mismatch Asymmetry. This effect is asymmetric, in which sentences with a passive antecedent and active ellipsis clause are generally judged as more acceptable than sentences with an active antecedent and passive ellipsis clause.

There are two leading theories in the cause of Mismatch Asymmetry in sentences with verb ellipses. The first theory is the Recycling Hypothesis. This hypothesis proposes that passive clauses are more prone to being misremembered as active clauses than vice versa. This creates an “illusion of grammaticality” for passive-active mismatches but not active-passive mismatches. The second theory is the Passive Ellipsis Clause Penalty (PECP). In this theory, passive ellipsis clauses are rated as less acceptable than active ellipsis clauses regardless of any mismatch between the ellipsis clause and its antecedent.

Poppels & Kehler Experiment 2 found that sentences with a passive ellipsis clause are judged as less acceptable than those with a passive antecedent, which provided support for the PECP hypothesis. However, they concluded that a more definitive experiment needed to be undertaken in order to provide more substantial evidence for the PECP explanation for Mismatch Asymmetry. Experiment 3 tests the RH vs. PECP hypotheses for Mismatch Asymmetry by testing a condition where RH and PECP would predict opposite outcomes. In this experiment, the ellipsis clause is the first clause of the sentence and the antecedent is the second; in Experiments 1 and 2, the antecedent clause was first and the ellipsis clause was second.

RH predicts that a sentence with a passive ellipsis first clause and active antecedent second clause should be *more acceptable* than an active ellipsis first clause and passive antecedent second clause. This is because the first clause – the ellipsis clause in this case – is the clause that is subject to misremembering. The passive clause will be misremembered as active, so the sentence will seem like $[A \leftarrow A]$, which is considered acceptable. If the sentence was $[A \leftarrow P]$, the P won't be misremembered as it is the second clause, and the sentence will seem less acceptable due to the mismatch.

PECP predicts the exact opposite as RH. It predicts that a sentence with a passive ellipsis first clause and active antecedent second clause should be *less acceptable* than an active ellipsis first clause and passive antecedent second clause. This is due to its general proposed penalty for passive ellipsis clauses.

The study presented in this paper attempts to replicate Poppels & Kehler Experiment 3 with a sample size of eighteen native English speakers. The findings were consistent with the original paper. There was a significant mismatch effect, a significant effect of the ellipsis voice in which passive ellipsis clauses were rated lower than active ellipsis clauses, and no interaction between the two main effects of mismatch and ellipsis clause voice.

2 Experiment

Experiment 2 performed a more definitive test as to whether the Mismatch Asymmetry seen in Experiments 1 and 2 was caused by RH or PECP. The experiment compared acceptability ratings between $[P \leftarrow P]$, $[P \leftarrow A]$, $[A \leftarrow A]$, and $[A \leftarrow P]$ sentences, in which the RH and PECP theories predicted opposite outcomes. The goal of this study was to replicate the results of Poppels & Kehler.

2.2 Methods

2.2.1 Design and Materials

This experiment utilized a 2×2 cross design. The two factors were mismatch (match vs. mismatch) and the voice of the ellipsis clause (passive vs. active). There were four conditions per item. Below is a full example:

Condition	Sentence
$[A \leftarrow A]$	Before her mother did, Mary's father scolded her.
$[P \leftarrow P]$	Before her sister was, Mary was scolded by her father.
$[P \leftarrow A]$	Before her sister was, Mary's father scolded her.
$[A \leftarrow P]$	Before her mother did, Mary was scolded by her father.

Participants were presented with one sentence at a time and asked to rank the grammatical acceptability of the sentence on a five-point scale. 1 was completely unacceptable and 5 was completely acceptable. Participants were instructed to rate each sentence on its grammatical acceptability and ignore its plausibility.

Each participant ranked 24 experimental items and 48 filler items. These filler items included upper-bound elliptical and lower-bound elliptical items, which were clearly acceptable and unacceptable sentences, respectively. Their ratings for these items were used to determine the upper and lower acceptability ratings for Figure 1. The experiment was designed with a Latin Cross in which there were four groups of participants ($A \rightarrow D$). Each group saw a different set of experimental trials but saw the same amount of each condition type. Experimental and filler trials were presented in a random order to each participant.

2.2.2 Procedure

The experiment was designed on PCIBex and data was collected through PCIBex. Participants received instructions to rank sentences on a scale of 1-5 based on how grammatically acceptable they were as sentences in the English language. Before the experiment began, participants saw practice conditions in which they were asked to rate the acceptability of four sentences unrelated to the experimental manipulations. After submitting their rating, they received feedback and an explanation of the correct acceptability rating. Participants were reminded to rate sentences on grammatical acceptability alone instead of plausibility, and that there was no time limit for completing the experiment.

2.2.3 Participants

The eighteen participants ranged from age 18 to 80. Participants were recruited from Cornell University and also from the experimenter's adult relatives. The link was sent to people individually. The participants received no financial compensation for their time. PCIBex prevented participants who were under 18 or whose native language was not English from participating.

2.3 Predictions

The experiment tested the RH and PECP theories. The experiment was expected to confirm the PECP theory. It was predicted that there should be a mismatch effect in which $[P \leftarrow A]$ and $[A \leftarrow P]$ cases were judged as less grammatically acceptable than $[P \leftarrow P]$ and $[A \leftarrow A]$ cases. It was also predicted that there would be a main effect of the ellipsis clause voice. More specifically, $[P \leftarrow P]$ cases would be less acceptable than $[A \leftarrow A]$ cases, and $[P \leftarrow A]$ mismatch cases would be less grammatically acceptable than $[A \leftarrow P]$ mismatch cases. The last prediction is that the same decrease in acceptability would be observed between $[A \leftarrow A]$ and $[P \leftarrow A]$ cases, and $[A \leftarrow P]$ and $[P \leftarrow P]$ cases. This would indicate that there is no interaction between the main effects of mismatch and ellipsis clause voice.

2.4 Results

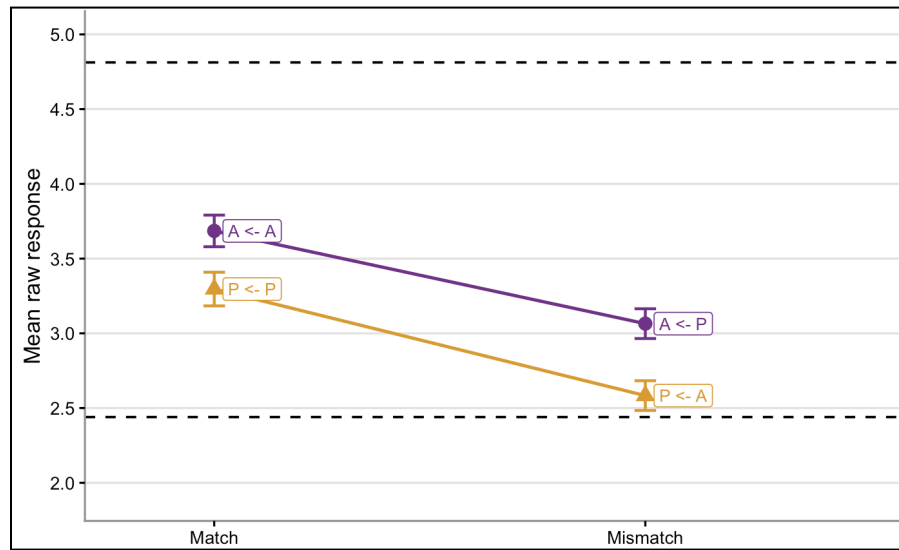


Figure 1. Results from Experiment 3 replication. Error bars reflect standard errors, dashed lines represent mean acceptability ratings of acceptable and unacceptable filler trials.

The data was cleaned by assigning a unique participant number to each participant based on their unique IP address. String response values were also converted to numeric values. The mean raw responses were sorted by sentence type (mismatch, voices of the clauses) and plotted in Figure 1.

The horizontal dashed lines in Figure 1 were generated by calculating the mean acceptability ratings for acceptable and unacceptable filler trials. This was done to provide boundaries that represented the total range of acceptability ratings that the participants were using.

The slopes of the lines in Figure 1 were also calculated to determine if the slopes were equal. The slope for the active ellipsis (purple line) condition was -0.620. The slope for the passive ellipsis (orange line) condition was -0.713.

An lmerTest was used to further determine the significance of the different main effects. The main effect of mismatch was determined to be highly significant ($\beta = -0.33$, $t = -8.44$, $p < .001$) and showed that mismatched conditions were rated lower than matched conditions. The effect of the ellipsis clause voice was also highly significant ($\beta = -0.23$, $t = -5.61$, $p < .001$) and showed that passive ellipsis clauses were rated lower than active ellipsis clauses. The interaction between the main effects of mismatch and ellipsis clause voice was determined to not be statistically significant ($\beta = -0.02$, $t = -0.59$, $p = .558$).

2.5 Discussion

The results of the replication study were consistent with the results from Poppels & Kehler. Figure 1 is very similar to the corresponding figure in their 2019 study. In both figures, the acceptability ratings for the experimental trials are much lower than the upper-bound dashed line. The slopes of the

lines are roughly parallel in both figures, and the passive ellipsis clause cases are rated lower than the active ellipsis clause cases. One noticeable difference is the higher overall ratings of sentences in this study versus Poppels & Kehler (about 0.5 higher) despite using the same stimuli. It is likely that this difference is due to the smaller sample size in this replication study (18 vs. 28 participants).

It is important to note that the slopes in Figure 1 were not exactly equal. They differed by 0.1, which does suggest some main effect of interaction, but could also be attributed to the small sample size. This is especially true when considering the results of the lmerTest in conjunction with Figure 1. Some problems were also encountered with the lmerTest. An lmerTest with random slopes generated an error, which was likely due to a sample size that was too small for the analysis. This experiment therefore used an lmerTest with random intercepts only, but still saw comparable results to the original study.

The results of the study confirmed all of the predictions and supported the PECP hypothesis. The mismatch main effect showed that mismatched conditions were rated significantly lower than matched conditions. Passive ellipsis clause voices were also ranked significantly lower than active ellipsis clause voices, providing strong evidence for the PECP hypothesis. The RH theory, by contrast, predicted that passive ellipsis clauses should be ranked *higher* than active ellipsis clauses. This experiment therefore contradicts the RH theory and suggests that PECP drives Mismatch Asymmetry. There was also no significant effect of interaction between the main effects of mismatch and ellipsis clause voice, which again supported the predictions made.

3 Conclusion

This paper sought to replicate the main results from Poppels & Kehler (2019) Experiment 3, in which they tested the RH versus PECP theories for Mismatch Asymmetry in a condition in which the two theories would predict opposite outcomes. Eighteen native English speakers completed an online acceptability judgement task in which they rated the grammatical acceptability of sentences with an ellipsis first clause and antecedent second clause. It was found that there was a significant mismatch effect in which sentences with mismatch were rated lower than sentences without. A significant ellipsis clause voice effect was also found that was consistent with the PECP theory. No significant interaction between the two main effects was observed. This study provided evidence for the PECP theory and evidence against the RH theory for Mismatch Asymmetry.

4 References

Poppels, Till and Andrew Kehler. 2019. Reconsidering asymmetries in voice-mismatched VP-ellipsis. *Glossa: a journal of general linguistics* 4(1): 60. 1–22. DOI: <https://doi.org/10.5334/gjgl.738>